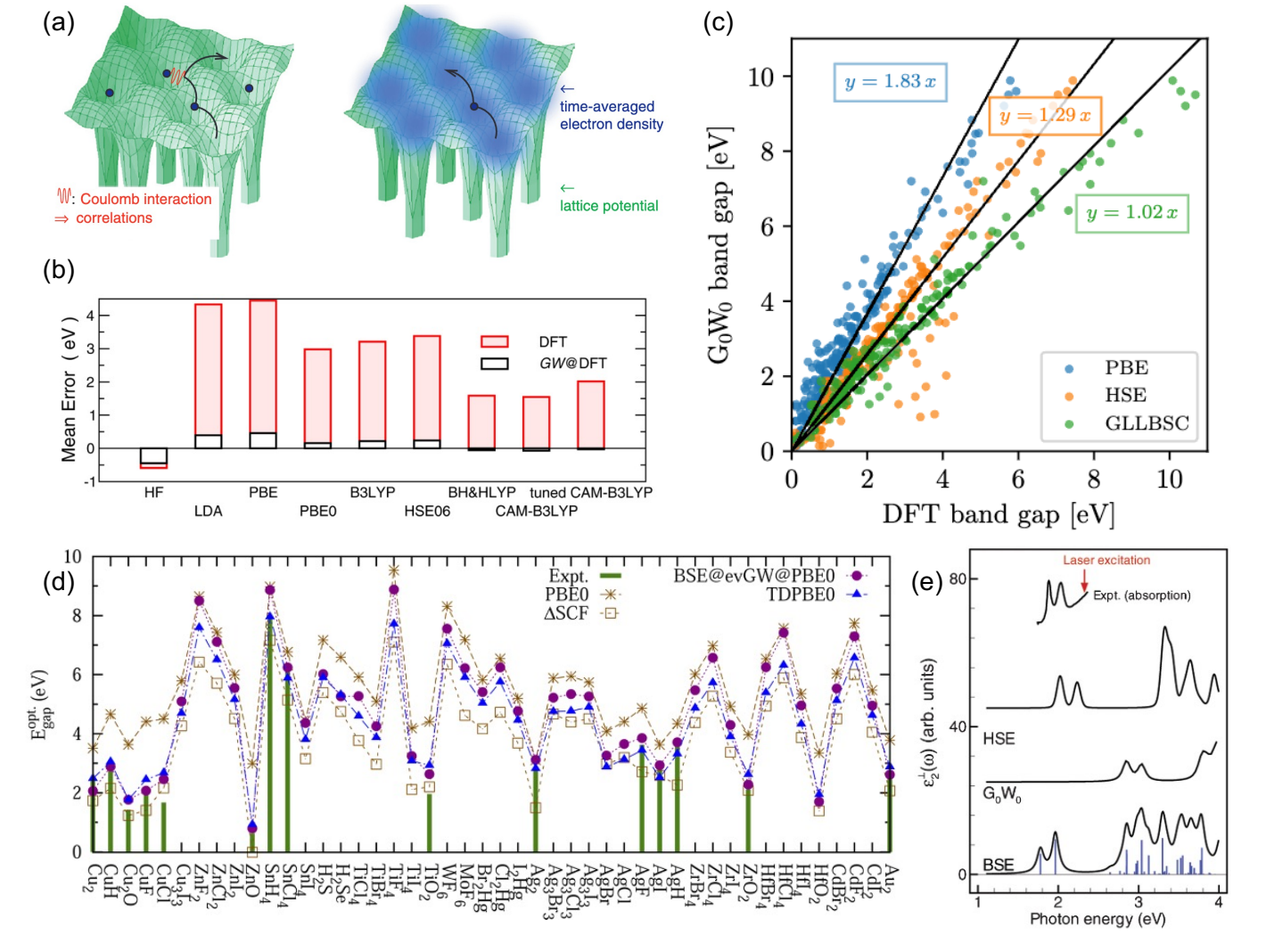

(a)
: Coulomb interaction
⇒ correlations
time-averaged
electron density
lattice potential
(b)
Mean Error (eV)
DFT
GW@DFT
HF
LDA
PBE
PBE0
B3LYP
HSE06
BH&HLYP
CAM-B3LYP
tuned CAM-B3LYP
(c)
$y = 1.83\,x$
$y = 1.29\,x$
$y = 1.02\,x$
PBE
HSE
GLLBSC
G_0W_0 band gap [eV]
DFT band gap [eV]
(d)
Expt.
PBE0
ΔSCF
BSE@evGW@PBE0
TDPBE0
$E_{gap}^{opt.}$ (eV)
(e)
Laser excitation
Expt. (absorption)
HSE
G_0W_0
BSE
$\varepsilon_2^{\perp}(\omega)$ (arb. units)
Photon energy (eV)